Venkata Lakshmi Parimala Pasupuleti

Income Fraud Detection & Risk Analytics Engineer · Python · SQL · AWS · ML Anomaly Detection · LIHTC / HCV / HUD · Document Forensics

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SUMMARY

Fraud detection and risk analytics engineer with **7+ years as the primary income fraud detection layer at Entrata** - one of the nation's largest multifamily SaaS platforms (100K+ monthly active housing units). Paystubs represent 69% of fraudulent documents in 2026 (Snappt Q3 report); template farms account for 41% of classified fraud cases. **Parimala was reviewing these exact documents manually at scale before automated systems existed** - detecting altered pay stubs, falsified bank statements, household-composition misrepresentation, and asset concealment across LIHTC, HCV, and HUD income certifications. Encoded that fraud judgment into **automated detection pipelines eliminating 80% of review workload with zero compliance violations** across 7+ years of annual recertification cycles. Unique differentiator: **recertification drift detection** - identifying fraud patterns that emerge between initial application and annual renewal, a gap neither Snappt nor Truework currently solve well. At Intuit, builds **ML fraud signal pipelines** (scikit-learn, XGBoost, Isolation Forest) and **AI-powered document verification** with deterministic rule engines ensuring LLMs surface evidence while fraud/eligibility decisions route to rule-based logic. Built **Leasel**: an agentic fraud detection and compliance audit pipeline (93.6% precision, 1,240 test cases). M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026).

CORE SKILLS

Fraud Detection & Risk: First-party income fraud, synthetic identity, document forensics (pay stubs, bank statements), asset concealment, household-composition fraud, coordinated fraud ring detection, recertification drift, signal gating, rule engines

Analytics & ML: Python, SQL, scikit-learn, XGBoost, Isolation Forest, Random Forest, Logistic Regression, SHAP explainability, NumPy, Pandas, feature engineering, real-time fraud signal pipelines

Compliance & Governance: LIHTC / Section 42, HCV / Section 8, HUD income limits & UIV, PHI / HIPAA, audit pipelines, eligibility rule engines, data lineage, access governance, SAR documentation

Cloud & Data Infrastructure: AWS (S3, Lambda, Glue, RDS, SQS, EC2), ETL pipelines, PostgreSQL, MongoDB, data validation & reconciliation, Docker, Kubernetes, GitHub Actions CI/CD, Terraform

EXPERIENCE

Intuit

Jun 2026 -- Present

Software Engineer - Financial Fraud Risk & ML Detection Pipelines - Virginia, USA

- Build **ML fraud signal detection pipelines** (scikit-learn, XGBoost, Isolation Forest): feature engineering on financial transaction and document data, model training, validation, and prediction APIs deployed on AWS - applying the same anomaly detection discipline used in income fraud to fintech financial data.
- Develop **AI-powered document verification workflows** with deterministic rule engines: LLM layer surfaces fraud evidence and inconsistencies; rule-based signal gates make all final fraud and eligibility decisions - zero AI false-positive risk on regulated financial outcomes.
- Implement **data governance and audit infrastructure** for PHI/regulated data: PII detection and redaction, access controls, and complete audit trails meeting compliance requirements for financial fraud investigation workflows.
- Build **fraud detection evaluation harnesses** integrated into CI/CD (GitHub Actions + golden datasets) - catching detection regressions before production release; SHAP-based model explainability for fraud analyst review.

Entrata - Multifamily & Affordable Housing SaaS

Jan 2017 -- Jul 2024

Software Engineer → Senior Software Engineer - Income Fraud Detection & Compliance Analytics - 100K+ monthly active housing units

- **Primary fraud detection layer** for affordable housing income certifications at scale: reviewed millions of income documents applying document forensics to detect altered pay stubs (69% of fraud cases per Snappt Q3 2026), falsified bank statements, and fabricated employment records - within LIHTC/Section 42, HCV, and HUD eligibility constraints requiring uniform, document-driven verification.
- Detected all four major fraud types end-to-end: **first-party income misrepresentation** (understating/overstating income), **synthetic identity indicators** (document-identity inconsistencies), **asset concealment** (hidden accounts to stay under Section 42 limits), and **household-composition fraud** (unreported occupants to maximize subsidy).
- Owned **recertification drift detection** - annual income recertification cycles where fraud patterns emerge between initial application and renewal; cross-certified household data against prior year declarations to flag income and asset discrepancies. This recurring detection gap is still largely manual across the industry.
- Encoded 7 years of fraud judgment into **automated detection pipelines** (Python, SQL, PostgreSQL): deterministic rule engines, eligibility signal gates, and anomaly flags processing 1.6M+ HUD/Tax Credit records annually - eliminating 80% of manual review workload with zero compliance violations and full audit trails for HUD and Tax Credit compliance reviewers.

- Built **fraud analytics tooling** adopted across teams: SQL-optimized eligibility rule engines (200+ queries, 30% performance gain), fraud investigation dashboards reducing triage time by 40%, and AWS data infrastructure (S3, Lambda, RDS) with Libsodium-encrypted PHI pipelines (100% resident data encryption, zero security incidents).

PROJECTS

Leasel - Agentic Income Fraud Detection & Compliance Audit Pipeline

Python · LangGraph · FastAPI · Deterministic Rule Engine · Signal Gating · LIHTC / HUD · CI/CD Eval

Production-grade fraud detection platform for affordable housing income certifications. **Document forensics layer** extracts income, employment, asset, and household data from uploaded documents (OCR + structured parsing). **Deterministic fraud signal engine**: applies LIHTC/Section 42 eligibility rules, flags first-party income misrepresentation, asset concealment, and household-composition fraud - with explicit signal gates routing all regulated decisions to rule-based logic rather than LLM output. **Agentic case review**: 8-agent LangGraph investigation graph with human-in-the-loop escalation when confidence is below threshold. Full evaluation harness: 93.6% precision, 87.9% F1 across 1,240 test cases.

github.com/venkatalakshmiparimala/Lease_complaince_rag

CrashClarity - End-to-End Risk Analytics & Anomaly Detection Pipeline (104K Records)

Python · scikit-learn · Isolation Forest · Random Forest · SHAP · NumPy · Pandas · Dashboard

Full risk analytics pipeline: ingestion, feature engineering, **Isolation Forest anomaly detection**, ML risk scoring (Random Forest + Logistic Regression, 85%+ accuracy), and **SHAP-based model explainability** for analyst review. Interactive self-service dashboard for non-technical fraud reviewers. Demonstrates end-to-end fraud risk analytics delivery from raw data through production-ready scoring and explainable outputs.

github.com/venkatalakshmiparimala

EDUCATION

M.S. Data Analytics Engineering — George Mason University, Virginia — May 2026 | GPA 3.87

B.Tech Computer Science & Engineering — JNTU Anantapur — 2017